DATA COMBINATION

2026-03-31

Packages & Data

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.4.3
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
# data import
dat <- read.csv("C:/Users/hjuvr/OneDrive/KNCL RSRCH/finaldatafile.csv")
```

Creating final data for analyses

```
# Mapping subjects onto the college they are taught at
subject_to_college <- c(
  # Carlson
  "ACCT"="Carlson","FINA"="Carlson","MGMT"="Carlson","MKTG"="Carlson",
  "SCMC"="Carlson","HRIR"="Carlson","IBUS"="Carlson","BA"="Carlson",
  "IDSC"="Carlson","BLAW"="Carlson","ENTR"="Carlson","SCO"="Carlson","SMGT"="Carlson",

  # CBS
  "BIOL"="CBS","EEB"="CBS","MICB"="CBS","PMB"="CBS","NSCI"="CBS","BIOC"="CBS",

  # CDes
  "ADES"="CDes","ARCH"="CDes","DES"="CDes","GDES"="CDes","PDES"="CDes",
  "LA"="CDes","IDES"="CDes","DH"="CDes",

  # CEHD
  "EPSY"="CEHD","OLPD"="CEHD","CPSY"="CEHD","KIN"="CEHD",
  "AHS"="CEHD","CI"="CEHD","HSM"="CEHD","YOST"="CEHD",

  # CFANS
  "ANSC"="CFANS","APEC"="CFANS","AGRO"="CFANS","FNRM"="CFANS",
  "FSCN"="CFANS","HORT"="CFANS","PLPA"="CFANS","PLSC"="CFANS",
  "SOIL"="CFANS","ESPM"="CFANS","FW"="CFANS","CFAN"="CFANS",
  "BSE"="CFANS","ABUS"="CFANS","AECM"="CFANS","BBE"="CFANS",
  "FSOS"="CFANS","MORT"="CFANS",

  # CSE
  "CSCI"="CSE","MATH"="CSE","STAT"="CSE","PHYS"="CSE","CHEN"="CSE",
  "CEGE"="CSE","ME"="CSE","EE"="CSE","BMEN"="CSE","MATS"="CSE",
  "ESCI"="CSE","HSCI"="CSE","AEM"="CSE","IE"="CSE","INET"="CSE",

  # Nursing
  "NURS"="Nursing",

  # CLA
  "AAS"="CLA","AFRO"="CLA","AMST"="CLA","ANTH"="CLA","ARTH"="CLA","ARTS"="CLA",
  "CHIC"="CLA","COMM"="CLA","ECON"="CLA","ENGL"="CLA","ENGW"="CLA","FREN"="CLA",
  "GEOG"="CLA","GER"="CLA","GLOS"="CLA","GWSS"="CLA","HIST"="CLA","ITAL"="CLA",
  "JOUR"="CLA","LING"="CLA","MUS"="CLA","PHIL"="CLA","POL"="CLA","PSY"="CLA",
  "RELS"="CLA","RUSS"="CLA","SCAN"="CLA","SOC"="CLA","SPAN"="CLA","TH"="CLA",
  "WRIT"="CLA","ESL"="CLA","CSCL"="CLA","HSEM"="CLA","SLHS"="CLA",
  "AMES"="CLA","AMIN"="CLA","ARAB"="CLA","CHN"="CLA","DNCE"="CLA",
  "EAS"="CLA","GLBT"="CLA","HEBR"="CLA","JPN"="CLA","KOR"="CLA",
  "LAT"="CLA","MEST"="CLA","OJIB"="CLA","PORT"="CLA","URBS"="CLA","URDU"="CLA"
)

# Averaging SRT data over all SRT columns
dat <- dat %>%
  mutate(
    avg_srt = rowMeans(
      select(
        .,
        deeper_understanding,
```

```
        interest_stimulated,
        tech_effective,
        activities_supported_learning,
        effort_reasonable,
        grading_clear
      ),
      na.rm = TRUE
    )
  )

dat$college <- unname(subject_to_college[dat$subject])

# Create GPA variable
dat <- dat %>%
  mutate(
    grade_points = case_when(
      crse_grade_off == "A"  ~ 4.000,
      crse_grade_off == "A-" ~ 3.667,
      crse_grade_off == "B+" ~ 3.333,
      crse_grade_off == "B"  ~ 3.000,
      crse_grade_off == "B-" ~ 2.667,
      crse_grade_off == "C+" ~ 2.333,
      crse_grade_off == "C"  ~ 2.000,
      crse_grade_off == "C-" ~ 1.667,
      crse_grade_off == "D+" ~ 1.333,
      crse_grade_off == "D"  ~ 1.000,
      crse_grade_off == "F"  ~ 0.000,
      TRUE ~ NA_real_
    )
  )

# College-level summary function
run_analysis_college <- function(data) {

  data %>%
    group_by(college, full_name, term) %>%
    summarise(

      avg_grade = ifelse(
        sum(grade_hdcnt, na.rm = TRUE) > 0,
        weighted.mean(grade_points, grade_hdcnt, na.rm = TRUE),
        NA_real_
      ),

      total_hdcnt = sum(
        grade_hdcnt[crse_grade_off %in% c("A","A-","B+","B","B-",
                                          "C+","C","C-","D+","D","F")],
        na.rm = TRUE
      ),

      pct_A = ifelse(
        total_hdcnt > 0,
```

```
        sum(
          grade_hdcnt[crse_grade_off %in% c("A", "A-")],
          na.rm = TRUE
        ) / total_hdcnt,
        NA_real_
      ),

      pct_C_or_lower = ifelse(
        total_hdcnt > 0,
        sum(
          grade_hdcnt[crse_grade_off %in% c("C+","C","C-","D+","D","F")],
          na.rm = TRUE
        ) / total_hdcnt,
        NA_real_
      ),

      rating = mean(avg_srt, na.rm = TRUE),

      award = ifelse(all(is.na(award_received)), NA, max(award_received, na.rm = TRUE)),

      .groups = "drop"
    )
}

# Run analysis
college_data <- run_analysis_college(dat)
```

College filtering

```
# Filter for colleges
CLA <- college_data %>% filter(college == "CLA")
CSE <- college_data %>% filter(college == "CSE")
CBS <- college_data %>% filter(college == "CBS")
Carlson <- college_data %>% filter(college == "Carlson")
CEHD <- college_data %>% filter(college == "CEHD")
CFANS <- college_data %>% filter(college == "CFANS")
CDes <- college_data %>% filter(college == "CDes")
Nursing <- college_data %>% filter(college == "Nursing")
```

Creating unique course clusters

```
run_analysis_subject <- function(data) {

  data %>%

    mutate(subject_group = case_when(

      subject %in% c("MATH","STAT","CSCI","PHYS","CHEM","AEM","BMEN","CEGE","CHEN","EE","IE","M
E","BBE","ESCI","ESPM","GCD","BIOC","BIOL","MICB","NSCI","PLPA","PLSC","SOIL") ~ "Quant STEM",

      subject %in% c("AEM","BMEN","CEGE","CHEN","EE","IE","ME","CMGT","INET","MATS") ~ "Engineer
ing",

      subject %in% c("ACCT","FINA","MKTG","SCO","BA","IBUS","APEC","ECON","ABUS","HRIR","RM") ~
"Business/Econ",

      subject %in% c("ANTH","SOC","PSY","POL","GEOG","FSOS","URBS","PA","GWSS","HSCI","PUBH") ~
"Social Science",

      subject %in% c("HIST","ENGL","PHIL","RELS","AMST","CSCL","ARTH","AMIN","AFRO","AMES","JWS
T","MEST","CNES") ~ "Humanities",

      subject %in% c("ARTS","DNCE","TH","MUS","DES","GDES","ARCH","LA","IDES","PDES") ~ "Arts &
Design",

      subject %in% c("COMM","JOUR","WRIT","ENGW","CI","EPSY","HSEM","HCOL") ~ "Communication/Pro
fessional",

      subject %in% c("NURS","DH","SLHS","KIN","HSM","PUBH","FSOS") ~ "Health",

      subject %in% c("FREN","SPAN","GER","RUSS","CHN","JPN","KOR","ITAL","ARAB","PORT","LAT","AS
L","SWAH","HEBR","URDU","HNDI","DAKO","SMLI","OJIB","HMNG") ~ "Languages",

      TRUE ~ "Other"
    )) %>%

    group_by(subject_group, full_name, term) %>%
    summarise(

      avg_grade = ifelse(
        sum(grade_hdcnt, na.rm = TRUE) > 0,
        weighted.mean(grade_points, grade_hdcnt, na.rm = TRUE),
        NA_real_
      ),

      total_hdcnt = sum(
        grade_hdcnt[crse_grade_off %in% c("A","A-","B+","B","B-",
                                          "C+","C","C-","D+","D","F")],
        na.rm = TRUE
      ),

      pct_A = ifelse(
        total_hdcnt > 0,
```

```
        sum(grade_hdcnt[crse_grade_off %in% c("A","A-")]) / total_hdcnt,
        NA_real_
      ),

      pct_C_or_lower = ifelse(
        total_hdcnt > 0,
        sum(grade_hdcnt[crse_grade_off %in% c("C+","C","C-","D+","D","F")]) / total_hdcnt,
        NA_real_
      ),

      rating = mean(avg_srt, na.rm = TRUE),

      award = ifelse(all(is.na(award_received)), NA,
                     max(award_received, na.rm = TRUE)),

      .groups = "drop"
    )
}

subject_data <- run_analysis_subject(dat)

Quant_STEM <- subject_data %>% filter(subject_group == "Quant STEM")
Engineering <- subject_data %>% filter(subject_group == "Engineering")
Business <- subject_data %>% filter(subject_group == "Business/Econ")
Social_Sci <- subject_data %>% filter(subject_group == "Social Science")
Humanities <- subject_data %>% filter(subject_group == "Humanities")
Arts <- subject_data %>% filter(subject_group == "Arts & Design")
Comm_Prof <- subject_data %>% filter(subject_group == "Communication/Professional")
Health <- subject_data %>% filter(subject_group == "Health")
Languages <- subject_data %>% filter(subject_group == "Languages")
Other <- subject_data %>% filter(subject_group == "Other")
```

Correlation tests

```
#avg grade vs SRT
cor.test(college_data$avg_grade, college_data$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  college_data$avg_grade and college_data$rating
## t = 22.208, df = 5844, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.2551611 0.3024436
## sample estimates:
##       cor
## 0.2789714
```

```
#percent of A's vs SRT
cor.test(college_data$pct_A, college_data$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  college_data$pct_A and college_data$rating
## t = 24.449, df = 5844, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.2811829 0.3276983
## sample estimates:
##       cor
## 0.3046222
```

```
#percent of C's or lower vs SRT
cor.test(college_data$pct_C_or_lower, college_data$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  college_data$pct_C_or_lower and college_data$rating
## t = -16.011, df = 5844, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.2294293 -0.1803124
## sample estimates:
##        cor
## -0.2049999
```

```
# CLA Grades vs SRTs
cor(CLA$avg_grade, CLA$rating, use = "complete.obs")
```

```
## [1] 0.296331
```

```
cor(CLA$pct_A, CLA$rating, use = "complete.obs")
```

```
## [1] 0.3156185
```

```
cor(CLA$pct_C_or_lower, CLA$rating, use = "complete.obs")
```

```
## [1] -0.2151474
```

```
# CSE Grades vs SRTs
cor(CSE$avg_grade, CSE$rating, use = "complete.obs")
```

```
## [1] 0.1717906
```

```
cor(CSE$pct_A, CSE$rating, use = "complete.obs")
```

```
## [1] 0.2412553
```

```
cor(CSE$pct_C_or_lower, CSE$rating, use = "complete.obs")
```

```
## [1] -0.1029265
```

```
# CBS Grades vs SRTs
cor(CBS$avg_grade, CBS$rating, use = "complete.obs")
```

```
## [1] 0.3193447
```

```
cor(CBS$pct_A, CBS$rating, use = "complete.obs")
```

```
## [1] 0.3096638
```

```
cor(CBS$pct_C_or_lower, CBS$rating, use = "complete.obs")
```

```
## [1] -0.2822985
```

```
# Carlson Grades vs SRTs
cor(Carlson$avg_grade, Carlson$rating, use = "complete.obs")
```

```
## [1] 0.176753
```

```
cor(Carlson$pct_A, Carlson$rating, use = "complete.obs")
```

```
## [1] 0.1806721
```

```
cor(Carlson$pct_C_or_lower, Carlson$rating, use = "complete.obs")
```

```
## [1] -0.1216535
```

```
# CEHD Grades vs SRTs
cor(CEHD$avg_grade, CEHD$rating, use = "complete.obs")
```

```
## [1] 0.2860278
```

```
cor(CEHD$pct_A, CEHD$rating, use = "complete.obs")
```

```
## [1] 0.300906
```

```
cor(CEHD$pct_C_or_lower, CEHD$rating, use = "complete.obs")
```

```
## [1] -0.2365096
```

```
# Awards
```

```
# CFANS Grades vs SRTs
cor(CFANS$avg_grade, CFANS$rating, use = "complete.obs")
```

```
## [1] 0.2115451
```

```
cor(CFANS$pct_A, CFANS$rating, use = "complete.obs")
```

```
## [1] 0.2251926
```

```
cor(CFANS$pct_C_or_lower, CFANS$rating, use = "complete.obs")
```

```
## [1] -0.153124
```

```
# CDes Grades vs SRTs
cor(CDes$avg_grade, CDes$rating, use = "complete.obs")
```

```
## [1] 0.2116947
```

```
cor(CDes$pct_A, CDes$rating, use = "complete.obs")
```

```
## [1] 0.1914262
```

```
cor(CDes$pct_C_or_lower, CDes$rating, use = "complete.obs")
```

```
## [1] -0.123798
```

```
award_summary <- college_data %>%
  filter(!is.na(award)) %>%
  group_by(award) %>%
  summarise(
    mean_avg_grade = mean(avg_grade, na.rm = TRUE),
    sd_avg_grade   = sd(avg_grade, na.rm = TRUE),

    mean_pct_A = mean(pct_A, na.rm = TRUE),
    sd_pct_A   = sd(pct_A, na.rm = TRUE),

    mean_pct_C_or_lower = mean(pct_C_or_lower, na.rm = TRUE),
    sd_pct_C_or_lower   = sd(pct_C_or_lower, na.rm = TRUE),

    mean_rating = mean(rating, na.rm = TRUE),
    sd_rating   = sd(rating, na.rm = TRUE),

    n = n()
  )

award_summary
```

```
## # A tibble: 2 × 10
##   award mean_avg_grade sd_avg_grade mean_pct_A sd_pct_A mean_pct_C_or_lower
##   <int>          <dbl>        <dbl>      <dbl>    <dbl>               <dbl>
## 1     0           3.50        0.346      0.672    0.220              0.0968
## 2     1           3.57        0.330      0.720    0.215              0.0856
## # ℹ 4 more variables: sd_pct_C_or_lower <dbl>, mean_rating <dbl>,
## #   sd_rating <dbl>, n <int>
```

d values for grades and teaching awards

```
g1 <- college_data$avg_grade[college_data$award == 1]
g0 <- college_data$avg_grade[college_data$award == 0]

m1 <- mean(g1, na.rm = TRUE)
m0 <- mean(g0, na.rm = TRUE)

s1 <- sd(g1, na.rm = TRUE)
s0 <- sd(g0, na.rm = TRUE)

n1 <- sum(!is.na(g1))
n0 <- sum(!is.na(g0))

sp <- sqrt(((n1 - 1)*s1^2 + (n0 - 1)*s0^2) / (n1 + n0 - 2))

d <- (m1 - m0) / sp
d
```

```
## [1] 0.2290136
```

```
g1 <- college_data$pct_A[college_data$award == 1]
g0 <- college_data$pct_A[college_data$award == 0]

m1 <- mean(g1, na.rm = TRUE)
m0 <- mean(g0, na.rm = TRUE)

s1 <- sd(g1, na.rm = TRUE)
s0 <- sd(g0, na.rm = TRUE)

n1 <- sum(!is.na(g1))
n0 <- sum(!is.na(g0))

sp <- sqrt(((n1 - 1)*s1^2 + (n0 - 1)*s0^2) / (n1 + n0 - 2))

d_pct_A <- (m1 - m0) / sp
d_pct_A
```

```
## [1] 0.220065
```

```
g1 <- college_data$pct_C_or_lower[college_data$award == 1]
g0 <- college_data$pct_C_or_lower[college_data$award == 0]

m1 <- mean(g1, na.rm = TRUE)
m0 <- mean(g0, na.rm = TRUE)

s1 <- sd(g1, na.rm = TRUE)
s0 <- sd(g0, na.rm = TRUE)

n1 <- sum(!is.na(g1))
n0 <- sum(!is.na(g0))

sp <- sqrt(((n1 - 1)*s1^2 + (n0 - 1)*s0^2) / (n1 + n0 - 2))

d_pct_C_or_lower <- (m1 - m0) / sp
d_pct_C_or_lower
```

```
## [1] -0.1040416
```

T-tests for pooled data

```
t_pct_A <- t.test(pct_A ~ award, data = college_data)
t_pct_A
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_A by award
## t = -2.0991, df = 89.818, p-value = 0.03861
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.094189586 -0.002590839
## sample estimates:
## mean in group 0 mean in group 1
##       0.6715145       0.7199047
```

```
t_avg <- t.test(avg_grade ~ award, data = college_data)
t_avg
```

```
##
##  Welch Two Sample t-test
##
## data:  avg_grade by award
## t = -2.2367, df = 89.96, p-value = 0.02778
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.149722407 -0.008862988
## sample estimates:
## mean in group 0 mean in group 1
##        3.495035        3.574328
```

```
t_pct_C <- t.test(pct_C_or_lower ~award, data = college_data)
t_pct_C
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_C_or_lower by award
## t = 0.98033, df = 89.747, p-value = 0.3296
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.01145397  0.03376794
## sample estimates:
## mean in group 0 mean in group 1
##      0.09679027      0.08563329
```

Now T-test's for college analyses :)

```
t.test(avg_grade ~ award, data = CLA)
```

```
##
##  Welch Two Sample t-test
##
## data:  avg_grade by award
## t = -4.2091, df = 50.145, p-value = 0.0001061
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.25133097 -0.08895752
## sample estimates:
## mean in group 0 mean in group 1
##        3.515444        3.685588
```

```
t.test(avg_grade ~ award, data = CSE)
```

```
##
##  Welch Two Sample t-test
##
## data:  avg_grade by award
## t = 0.51521, df = 8.1808, p-value = 0.62
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.3031116  0.4783866
## sample estimates:
## mean in group 0 mean in group 1
##        3.267782        3.180144
```

```
t.test(avg_grade ~ award, data = CBS)
```

```
##
##  Welch Two Sample t-test
##
## data:  avg_grade by award
## t = -2.1392, df = 2.8944, p-value = 0.1252
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.4867121  0.1002575
## sample estimates:
## mean in group 0 mean in group 1
##        3.290034        3.483261
```

```
t.test(avg_grade ~ award, data = CFANS)
```

```
##
##  Welch Two Sample t-test
##
## data:  avg_grade by award
## t = 3.0367, df = 10.609, p-value = 0.01176
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  0.0388974 0.2471804
## sample estimates:
## mean in group 0 mean in group 1
##        3.402620        3.259581
```

```
t.test(avg_grade ~ award, data = CDes)
```

```
##
##  Welch Two Sample t-test
##
## data:  avg_grade by award
## t = 0.64172, df = 1.046, p-value = 0.6329
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -1.339060  1.497947
## sample estimates:
## mean in group 0 mean in group 1
##        3.552317        3.472874
```

```
t.test(pct_A ~ award, data = CLA)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_A by award
## t = -5.2259, df = 50.505, p-value = 3.316e-06
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.16765108 -0.07457577
## sample estimates:
## mean in group 0 mean in group 1
##       0.6945789       0.8156923
```

```
t.test(pct_A ~ award, data = CSE)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_A by award
## t = 0.298, df = 8.1737, p-value = 0.7731
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.2079731  0.2699637
## sample estimates:
## mean in group 0 mean in group 1
##       0.5094209       0.4784257
```

```
t.test(pct_A ~ award, data = CBS)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_A by award
## t = -0.75574, df = 3.3873, p-value = 0.4989
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.15453171  0.09211112
## sample estimates:
## mean in group 0 mean in group 1
##       0.5460933       0.5773035
```

```
t.test(pct_A ~ award, data = CFANS)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_A by award
## t = 3.4888, df = 10.014, p-value = 0.005823
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  0.04793722 0.21730081
## sample estimates:
## mean in group 0 mean in group 1
##       0.5998495       0.4672305
```

```
t.test(pct_A ~ award, data = CDes)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_A by award
## t = 0.90688, df = 1.0281, p-value = 0.5277
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -1.344751  1.566360
## sample estimates:
## mean in group 0 mean in group 1
##       0.6680390       0.5572344
```

```
t.test(pct_C_or_lower ~ award, data = CLA)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_C_or_lower by award
## t = 3.0903, df = 50.68, p-value = 0.003244
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  0.01156628 0.05447696
## sample estimates:
## mean in group 0 mean in group 1
##      0.09117631      0.05815468
```

```
t.test(pct_C_or_lower ~ award, data = CSE)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_C_or_lower by award
## t = -1.5299, df = 8.1728, p-value = 0.1638
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.24262146  0.04866404
## sample estimates:
## mean in group 0 mean in group 1
##       0.1595265       0.2565052
```

```
t.test(pct_C_or_lower ~ award, data = CBS)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_C_or_lower by award
## t = 3.922, df = 4.946, p-value = 0.0114
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  0.03034649 0.14690033
## sample estimates:
## mean in group 0 mean in group 1
##      0.17228173      0.08365832
```

```
t.test(pct_C_or_lower ~ award, data = CFANS)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_C_or_lower by award
## t = -1.136, df = 9.8137, p-value = 0.2829
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.07356270  0.02396588
## sample estimates:
## mean in group 0 mean in group 1
##       0.1212946       0.1460930
```

```
t.test(pct_C_or_lower ~ award, data = CDes)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_C_or_lower by award
## t = 1.8364, df = 1.9525, p-value = 0.2108
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.02363487  0.05743381
## sample estimates:
## mean in group 0 mean in group 1
##      0.06039764      0.04349817
```

NOW FOR COURSE CLUSTERS

```
# Quant STEM
cor.test(Quant_STEM$avg_grade, Quant_STEM$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Quant_STEM$avg_grade and Quant_STEM$rating
## t = 5.9604, df = 671, p-value = 4.064e-09
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.151228 0.294820
## sample estimates:
##       cor
## 0.2242408
```

```
cor.test(Quant_STEM$pct_A, Quant_STEM$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Quant_STEM$pct_A and Quant_STEM$rating
## t = 7.3734, df = 671, p-value = 4.917e-13
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.2023831 0.3422653
## sample estimates:
##       cor
## 0.2737713
```

```
cor.test(Quant_STEM$pct_C_or_lower, Quant_STEM$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Quant_STEM$pct_C_or_lower and Quant_STEM$rating
## t = -4.0557, df = 671, p-value = 5.585e-05
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.22759951 -0.08004473
## sample estimates:
##        cor
## -0.1546846
```

```
# Engineering
cor.test(Engineering$avg_grade, Engineering$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Engineering$avg_grade and Engineering$rating
## t = 2.0192, df = 67, p-value = 0.04747
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.002996978 0.450642940
## sample estimates:
##       cor
## 0.2395078
```

```
cor.test(Engineering$pct_A, Engineering$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Engineering$pct_A and Engineering$rating
## t = 2.8386, df = 67, p-value = 0.005992
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.09861577 0.52371642
## sample estimates:
##      cor
## 0.327649
```

```
cor.test(Engineering$pct_C_or_lower, Engineering$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Engineering$pct_C_or_lower and Engineering$rating
## t = -1.154, df = 67, p-value = 0.2526
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.3642455  0.1003982
## sample estimates:
##        cor
## -0.1395997
```

```
# Business / Econ
cor.test(Business$avg_grade, Business$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Business$avg_grade and Business$rating
## t = 2.8785, df = 298, p-value = 0.004284
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.05220931 0.27264047
## sample estimates:
##       cor
## 0.1644777
```

```
cor.test(Business$pct_A, Business$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Business$pct_A and Business$rating
## t = 2.6391, df = 298, p-value = 0.00875
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.0385408 0.2599149
## sample estimates:
##       cor
## 0.1511221
```

```
cor.test(Business$pct_C_or_lower, Business$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Business$pct_C_or_lower and Business$rating
## t = -3.2982, df = 298, p-value = 0.001091
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.29464509 -0.07604102
## sample estimates:
##        cor
## -0.1876659
```

```
# Social Science
cor.test(Social_Sci$avg_grade, Social_Sci$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Social_Sci$avg_grade and Social_Sci$rating
## t = 7.6988, df = 992, p-value = 3.313e-14
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.1778922 0.2952663
## sample estimates:
##       cor
## 0.2374457
```

```
cor.test(Social_Sci$pct_A, Social_Sci$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Social_Sci$pct_A and Social_Sci$rating
## t = 7.994, df = 992, p-value = 3.616e-15
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.1866843 0.3035453
## sample estimates:
##       cor
## 0.2460086
```

```
cor.test(Social_Sci$pct_C_or_lower, Social_Sci$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Social_Sci$pct_C_or_lower and Social_Sci$rating
## t = -5.5423, df = 992, p-value = 3.825e-08
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.2329762 -0.1123374
## sample estimates:
##        cor
## -0.1733068
```

```
# Humanities
cor.test(Humanities$avg_grade, Humanities$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Humanities$avg_grade and Humanities$rating
## t = 9.432, df = 881, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.2417027 0.3615975
## sample estimates:
##       cor
## 0.3028478
```

```
cor.test(Humanities$pct_A, Humanities$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Humanities$pct_A and Humanities$rating
## t = 10.754, df = 881, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.2809788 0.3976759
## sample estimates:
##       cor
## 0.3406386
```

```
cor.test(Humanities$pct_C_or_lower, Humanities$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Humanities$pct_C_or_lower and Humanities$rating
## t = -5.5716, df = 881, p-value = 3.354e-08
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.2474522 -0.1199756
## sample estimates:
##        cor
## -0.1844897
```

```
# Arts & Design
cor.test(Arts$avg_grade, Arts$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Arts$avg_grade and Arts$rating
## t = 7.4517, df = 472, p-value = 4.432e-13
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.2414273 0.4027357
## sample estimates:
##       cor
## 0.3244383
```

```
cor.test(Arts$pct_A, Arts$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Arts$pct_A and Arts$rating
## t = 7.6362, df = 472, p-value = 1.253e-13
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.2489679 0.4094356
## sample estimates:
##      cor
## 0.331598
```

```
cor.test(Arts$pct_C_or_lower, Arts$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Arts$pct_C_or_lower and Arts$rating
## t = -4.9342, df = 472, p-value = 1.117e-06
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.3054486 -0.1340849
## sample estimates:
##        cor
## -0.2214759
```

```
# Communication / Professional
cor.test(Comm_Prof$avg_grade, Comm_Prof$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Comm_Prof$avg_grade and Comm_Prof$rating
## t = 8.1705, df = 568, p-value = 2.004e-15
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.2487982 0.3958792
## sample estimates:
##       cor
## 0.3242973
```

```
cor.test(Comm_Prof$pct_A, Comm_Prof$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Comm_Prof$pct_A and Comm_Prof$rating
## t = 8.2554, df = 568, p-value = 1.065e-15
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.2519565 0.3987169
## sample estimates:
##       cor
## 0.3273092
```

```
cor.test(Comm_Prof$pct_C_or_lower, Comm_Prof$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Comm_Prof$pct_C_or_lower and Comm_Prof$rating
## t = -6.127, df = 568, p-value = 1.675e-09
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.3244760 -0.1703434
## sample estimates:
##        cor
## -0.2489855
```

```
# Health
cor.test(Health$avg_grade, Health$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Health$avg_grade and Health$rating
## t = 4.5633, df = 242, p-value = 8.013e-06
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.1616036 0.3931645
## sample estimates:
##       cor
## 0.2814768
```

```
cor.test(Health$pct_A, Health$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Health$pct_A and Health$rating
## t = 4.8416, df = 242, p-value = 2.298e-06
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.1782333 0.4075437
## sample estimates:
##       cor
## 0.2971674
```

```
cor.test(Health$pct_C_or_lower, Health$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Health$pct_C_or_lower and Health$rating
## t = -3.3112, df = 242, p-value = 0.001071
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.32526754 -0.08481765
## sample estimates:
##        cor
## -0.2081859
```

```
# Languages
cor.test(Languages$avg_grade, Languages$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Languages$avg_grade and Languages$rating
## t = 4.0985, df = 358, p-value = 5.149e-05
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.1107630 0.3083165
## sample estimates:
##       cor
## 0.2117012
```

```
cor.test(Languages$pct_A, Languages$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Languages$pct_A and Languages$rating
## t = 5.0598, df = 358, p-value = 6.722e-07
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.1592302 0.3522949
## sample estimates:
##       cor
## 0.2583402
```

```
cor.test(Languages$pct_C_or_lower, Languages$rating, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  Languages$pct_C_or_lower and Languages$rating
## t = -2.2171, df = 358, p-value = 0.02724
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.21713256 -0.01317946
## sample estimates:
##        cor
## -0.1163827
```

```
# Quant STEM
t.test(avg_grade ~ award, data = Quant_STEM)
```

```
##
##  Welch Two Sample t-test
##
## data:  avg_grade by award
## t = -0.81299, df = 4.0339, p-value = 0.4615
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.7839967  0.4279446
## sample estimates:
## mean in group 0 mean in group 1
##        3.298028        3.476054
```

```
t.test(avg_grade ~ award, data = Engineering)
```

```
##
##  Welch Two Sample t-test
##
## data:  avg_grade by award
## t = 3.1106, df = 4.9594, p-value = 0.02682
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  0.06058712 0.64566549
## sample estimates:
## mean in group 0 mean in group 1
##        3.163383        2.810257
```

```
t.test(avg_grade ~ award, data = Business)
```

```
##
##  Welch Two Sample t-test
##
## data:  avg_grade by award
## t = 3.7046, df = 12.697, p-value = 0.002748
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  0.07583926 0.28928006
## sample estimates:
## mean in group 0 mean in group 1
##        3.442140        3.259581
```

```
t.test(avg_grade ~ award, data = Social_Sci)
```

```
##
##  Welch Two Sample t-test
##
## data:  avg_grade by award
## t = -5.1792, df = 22.04, p-value = 3.399e-05
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.2603822 -0.1114916
## sample estimates:
## mean in group 0 mean in group 1
##        3.478069        3.664006
```

```
t.test(avg_grade ~ award, data = Humanities)
```

```
##
##  Welch Two Sample t-test
##
## data:  avg_grade by award
## t = -4.9256, df = 33.188, p-value = 2.266e-05
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.3587852 -0.1490646
## sample estimates:
## mean in group 0 mean in group 1
##        3.470871        3.724796
```

```
t.test(avg_grade ~ award, data = Arts)
```

```
##
##  Welch Two Sample t-test
##
## data:  avg_grade by award
## t = -0.3718, df = 5.2593, p-value = 0.7246
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.2410018  0.1793063
## sample estimates:
## mean in group 0 mean in group 1
##        3.654991        3.685839
```

```
t.test(avg_grade ~ award, data = Comm_Prof)
```

```
##
##  Welch Two Sample t-test
##
## data:  avg_grade by award
## t = -0.62616, df = 1.0063, p-value = 0.6433
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -3.245825  2.936637
## sample estimates:
## mean in group 0 mean in group 1
##        3.565598        3.720192
```

```
t.test(avg_grade ~ award, data = Health)
```

```
##
##  Welch Two Sample t-test
##
## data:  avg_grade by award
## t = -0.5542, df = 2.3257, p-value = 0.6282
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.4289285  0.3190796
## sample estimates:
## mean in group 0 mean in group 1
##        3.432785        3.487709
```

```
t.test(avg_grade ~ award, data = Quant_STEM)
```

```
##
##  Welch Two Sample t-test
##
## data:  avg_grade by award
## t = -0.81299, df = 4.0339, p-value = 0.4615
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.7839967  0.4279446
## sample estimates:
## mean in group 0 mean in group 1
##        3.298028        3.476054
```

```
t.test(avg_grade ~ award, data = Engineering)
```

```
##
##  Welch Two Sample t-test
##
## data:  avg_grade by award
## t = 3.1106, df = 4.9594, p-value = 0.02682
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  0.06058712 0.64566549
## sample estimates:
## mean in group 0 mean in group 1
##        3.163383        2.810257
```

```
t.test(avg_grade ~ award, data = Business)
```

```
##
##  Welch Two Sample t-test
##
## data:  avg_grade by award
## t = 3.7046, df = 12.697, p-value = 0.002748
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  0.07583926 0.28928006
## sample estimates:
## mean in group 0 mean in group 1
##        3.442140        3.259581
```

```
t.test(avg_grade ~ award, data = Social_Sci)
```

```
##
##  Welch Two Sample t-test
##
## data:  avg_grade by award
## t = -5.1792, df = 22.04, p-value = 3.399e-05
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.2603822 -0.1114916
## sample estimates:
## mean in group 0 mean in group 1
##        3.478069        3.664006
```

```
t.test(avg_grade ~ award, data = Humanities)
```

```
##
##  Welch Two Sample t-test
##
## data:  avg_grade by award
## t = -4.9256, df = 33.188, p-value = 2.266e-05
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.3587852 -0.1490646
## sample estimates:
## mean in group 0 mean in group 1
##        3.470871        3.724796
```

```
t.test(avg_grade ~ award, data = Arts)
```

```
##
##  Welch Two Sample t-test
##
## data:  avg_grade by award
## t = -0.3718, df = 5.2593, p-value = 0.7246
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.2410018  0.1793063
## sample estimates:
## mean in group 0 mean in group 1
##        3.654991        3.685839
```

```
t.test(avg_grade ~ award, data = Comm_Prof)
```

```
##
##  Welch Two Sample t-test
##
## data:  avg_grade by award
## t = -0.62616, df = 1.0063, p-value = 0.6433
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -3.245825  2.936637
## sample estimates:
## mean in group 0 mean in group 1
##        3.565598        3.720192
```

```
t.test(avg_grade ~ award, data = Health)
```

```
##
##  Welch Two Sample t-test
##
## data:  avg_grade by award
## t = -0.5542, df = 2.3257, p-value = 0.6282
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.4289285  0.3190796
## sample estimates:
## mean in group 0 mean in group 1
##        3.432785        3.487709
```

```
t.test(pct_A ~ award, data = Quant_STEM)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_A by award
## t = -0.72886, df = 4.0263, p-value = 0.5062
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.5271177  0.3074624
## sample estimates:
## mean in group 0 mean in group 1
##       0.5319268       0.6417544
```

```
t.test(pct_A ~ award, data = Engineering)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_A by award
## t = 3.8708, df = 7.8288, p-value = 0.004937
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  0.07668864 0.30486323
## sample estimates:
## mean in group 0 mean in group 1
##       0.4650406       0.2742647
```

```
t.test(pct_A ~ award, data = Business)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_A by award
## t = 3.8714, df = 11.282, p-value = 0.002484
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  0.06568646 0.23757301
## sample estimates:
## mean in group 0 mean in group 1
##       0.6188602       0.4672305
```

```
t.test(pct_A ~ award, data = Social_Sci)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_A by award
## t = -4.7305, df = 22.171, p-value = 9.951e-05
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.14730093 -0.05753817
## sample estimates:
## mean in group 0 mean in group 1
##       0.6753141       0.7777336
```

```
t.test(pct_A ~ award, data = Humanities)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_A by award
## t = -5.201, df = 33.193, p-value = 1.004e-05
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.22113830 -0.09679704
## sample estimates:
## mean in group 0 mean in group 1
##       0.6728868       0.8318545
```

```
t.test(pct_A ~ award, data = Arts)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_A by award
## t = 0.11911, df = 5.1424, p-value = 0.9097
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.1897055  0.2083019
## sample estimates:
## mean in group 0 mean in group 1
##       0.7652129       0.7559146
```

```
t.test(pct_A ~ award, data = Comm_Prof)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_A by award
## t = -1.674, df = 1.0147, p-value = 0.3402
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -1.474272  1.120569
## sample estimates:
## mean in group 0 mean in group 1
##       0.7178853       0.8947368
```

```
t.test(pct_A ~ award, data = Health)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_A by award
## t = -1.6274, df = 3.1816, p-value = 0.1969
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.17375541  0.05367043
## sample estimates:
## mean in group 0 mean in group 1
##       0.6132884       0.6733309
```

```
t.test(pct_C_or_lower ~ award, data = Quant_STEM)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_C_or_lower by award
## t = 0.064331, df = 4.0324, p-value = 0.9518
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.2203098  0.2307951
## sample estimates:
## mean in group 0 mean in group 1
##       0.1541789       0.1489363
```

```
t.test(pct_C_or_lower ~ award, data = Engineering)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_C_or_lower by award
## t = -4.0416, df = 4.0374, p-value = 0.0153
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.32649659 -0.06115846
## sample estimates:
## mean in group 0 mean in group 1
##       0.1971389       0.3909664
```

```
t.test(pct_C_or_lower ~ award, data = Business)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_C_or_lower by award
## t = -2.0879, df = 10.948, p-value = 0.06098
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.096254242  0.002566726
## sample estimates:
## mean in group 0 mean in group 1
##      0.09924921      0.14609297
```

```
t.test(pct_C_or_lower ~ award, data = Social_Sci)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_C_or_lower by award
## t = 4.8234, df = 22.256, p-value = 7.854e-05
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  0.02804161 0.07029391
## sample estimates:
## mean in group 0 mean in group 1
##      0.10286964      0.05370188
```

```
t.test(pct_C_or_lower ~ award, data = Humanities)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_C_or_lower by award
## t = 3.3959, df = 34.164, p-value = 0.001749
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  0.01896631 0.07547161
## sample estimates:
## mean in group 0 mean in group 1
##      0.09684407      0.04962511
```

```
t.test(pct_C_or_lower ~ award, data = Arts)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_C_or_lower by award
## t = 3.4965, df = 6.9576, p-value = 0.01014
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  0.01029537 0.05347501
## sample estimates:
## mean in group 0 mean in group 1
##      0.05639055      0.02450536
```

```
t.test(pct_C_or_lower ~ award, data = Comm_Prof)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_C_or_lower by award
## t = 0.52687, df = 1.0104, p-value = 0.6906
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.6265389  0.6821426
## sample estimates:
## mean in group 0 mean in group 1
##      0.08043343      0.05263158
```

```
t.test(pct_C_or_lower ~ award, data = Health)
```

```
##
##  Welch Two Sample t-test
##
## data:  pct_C_or_lower by award
## t = 0.37265, df = 2.2935, p-value = 0.7411
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
0
## 95 percent confidence interval:
##  -0.1244590  0.1513989
## sample estimates:
## mean in group 0 mean in group 1
##      0.11076024      0.09729031
```